

Solution Requirements

Date	12 July 2026
Team ID	6a2fbf036b1ed0bd8fa789aa
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register and log in with email/password. On success the system issues a JWT access token, validated on every API call via JWT Authentication Middleware.	High
2	Authorization levels	Two roles: Borrower (manage own loans, request settlements, generate negotiation letters) and Admin (view aggregated data, manage lender configurations). Access is enforced via role-based access control.	High
3	External interfaces	Integrates with the Google Gemini API for AI-generated negotiation letters and strategies. Uses SQLite via SQLAlchemy ORM for persistence.	High
4	Transactions processing	Handles loan data submission, settlement request creation, and negotiation letter generation. Financial calculations (EMI ratio, debt stress score) are processed server-side and persisted to the database.	Medium
5	Reporting	The Financial Dashboard displays EMI ratio, monthly surplus, debt stress level, and settlement percentage. The AI History Module reports past negotiation letters and their outcomes.	Medium
6	Business rules, etc.	Settlement recommendations are calculated from overdue duration, outstanding balance, EMI, and monthly income. If the Gemini API is unavailable, a rule-based fallback ensures recommendations are still generated.	High
7	Compliance to laws or regulations	Borrower financial data must be handled per data privacy best practices, including encryption at rest/in transit and applicable regional financial data protection regulations.	High
8	Other	Support exporting negotiation letters as downloadable PDF/Word documents so borrowers can send them directly to their lender.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	API responses for financial calculations should return quickly; AI-generated negotiation letters should complete within an acceptable wait time.	Dashboard loads in < 2 seconds; negotiation letter generated in < 8 seconds.
2	Scalability	The system should support a growing number of borrowers and loan records without performance degradation, with the backend able to migrate from SQLite to a scalable RDBMS if needed.	Supports up to 5,000 concurrent borrower sessions.
3	Security & Data Privacy	Borrower financial data must be protected via encryption, secure password hashing, and JWT-based session security.	AES-256 encryption at rest, TLS 1.2+ in transit, bcrypt password hashing.
4	Reliability & Availability	The platform should remain available and gracefully fall back to rule-based logic when the Gemini API is unreachable.	99.5% uptime per month; fallback engages within 2 seconds of AI timeout.
5	Usability & Accessibility	The React.js frontend should be responsive across devices and easy to navigate for non-technical borrowers under financial stress.	Mobile-responsive design; complies with WCAG 2.1 AA standards.
6	Other	Codebase should be modular (FastAPI routers, SQLAlchemy models) to ease maintenance and allow deployment across cloud environments.	Deployable via Docker to any major cloud provider.